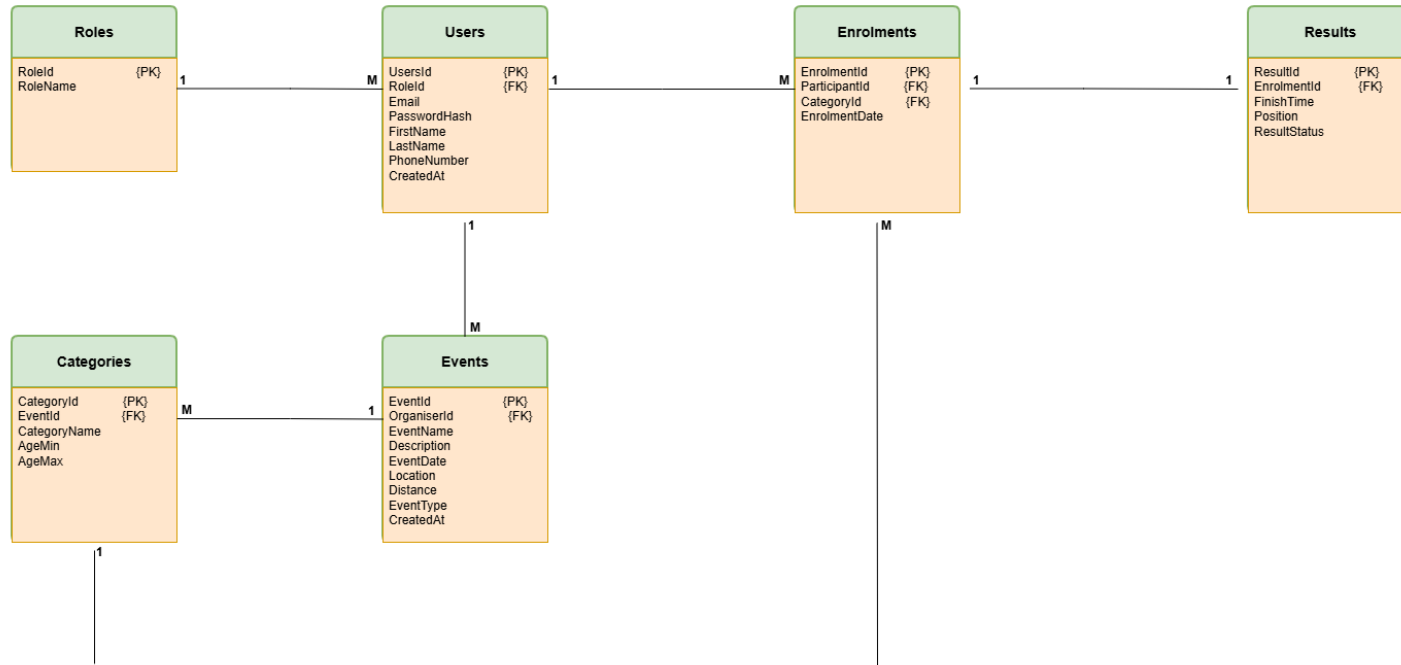


PROG6212 PART 1

STUDENT NUMBER: ST10478875

[Email address]

SECTION A – ERD SCREENSHOT BELOW



SECTION B

1.Authentication & Users

HTTP Method	Endpoint	Description	Access	Request Body	Response Status / Payload
POST	/api/auth/register	Register a new user account	Public	{"Email", "password", "firstName", "lastName", "phoneNumber", "RoleId"}	201 Created {"UserId", "email", "token"}
POST	/api/auth/login	Authenticate user and issue JWT	Public	{"email", "password"}	200 OK {"token", "UserId", "role"}
GET	/api/users/profile	Get current logged-in user profile	Authenticated	None	200 OK {"userId", "firstName", "lastName", "email", "phoneNumber"}

2. Events Management

HTTP Method	Endpoint	Description	Access	Request Body	Response Status / Payload
GET	/api/events	List all upcoming public events	Public	None	200 OK [{"eventId", "eventName", "location", "eventDate", "distance", "eventType"}]
GET	/api/events/{id}	Get full details for a single event	Public	None	200 OK {"eventId", "eventName", "description", "location", "distance", "categories": []}
POST	/api/events	Create a new race event	Organiser	{"eventName", "description", "eventDate", "location", "distance", "eventType"}	201 Created {"eventId", "eventName", "createdAt"}
PUT	/api/events/{id}	Update existing event details	Organiser	{ "eventName", "description", "eventDate", "location", "distance", "eventType" }	200 OK {"eventId", "updatedAt"}
DELETE	/api/events/{id}	Remove an event	Organiser	None	204 No Content

3. Categories Management

HTTP Method	Endpoint	Description	Access	Request Body	Response Status / Payload
GET	/api/events/{eventId}/categories	Get all categories for an event	Public	None	200 OK [{"categoryId", "categoryName", "ageMin", "ageMax"}]
POST	/api/events/{eventId}/categories	Add a new category to an event	Organiser	{ "categoryName", "ageMin", "ageMax" }	201 Created { "categoryId", "categoryName" }

4. Enrolments

HTTP Method	Endpoint	Description	Access	Request Body	Response Status / Payload
POST	/api/enrolments	Enrol participant into a category	Participant	{ "categoryId" }	201 Created { "enrolmentId", "participantId", "categoryId", "enrolmentDate" }
GET	/api/enrolments/my-enrolments	View user's own enrolments	Participant	None	200 OK

					[{"enrolmentId", "eventName", "categoryName", "enrolmentDate" }]
GET	/api/events/{eventId}/enrolments	View all entries for an event	Organiser	<i>None</i>	200 OK [{"enrolmentId", "participantName", "categoryName"}]

5. Race Results

HTTP Method	Endpoint	Description	Access	Request Body	Response Status / Payload
POST	/api/results	Capture race finish time and status	Organiser	{ "enrolmentId", "finishTime", "position", "resultStatus" }	201 Created { "resultId", "enrolmentId", "resultStatus" }
GET	/api/events/{eventId}/results	Get full official results for event	Public	<i>None</i>	200 OK [{"position", "participantName", "categoryName", "finishTime", "status"}]

SECTION C

CREATE DATABASE RaceDayDb;

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays a SQL query with two lines: `CREATE DATABASE RaceDayDb;` and `SELECT name FROM sys.databases;`. The bottom pane shows the results of the query, which is a list of databases. The database `RaceDayDb` is highlighted in blue, indicating it was successfully created. The interface also shows a status bar with "No issues found" and a message pane.

	name
1	master
2	tempdb
3	model
4	msdb
5	tea
6	CarDealerShip
7	Practice
8	RaceDayDb
9	Music


```
USE RaceDayDb;
```

```
-- DROP TABLES
```

```
IF OBJECT_ID('Results', 'U') IS NOT NULL  
    DROP TABLE Results;
```

```
IF OBJECT_ID('Enrolments', 'U') IS NOT NULL  
    DROP TABLE Enrolments;
```

```
IF OBJECT_ID('Categories', 'U') IS NOT NULL  
    DROP TABLE Categories;
```

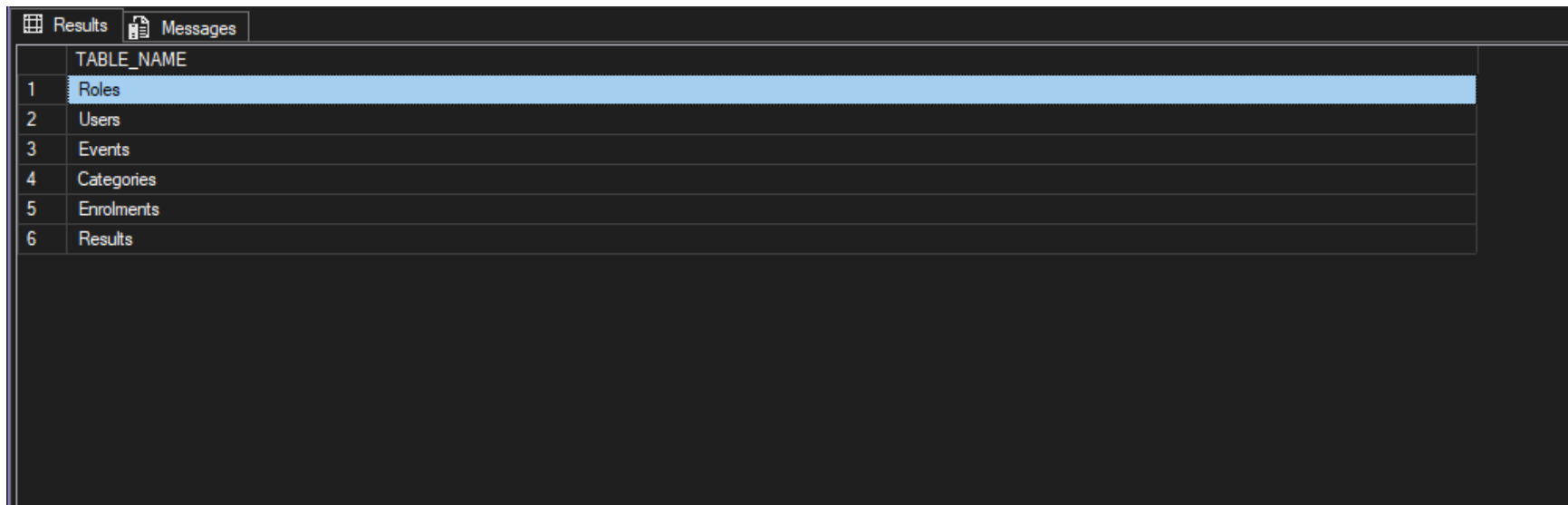
```
IF OBJECT_ID('Events', 'U') IS NOT NULL  
    DROP TABLE Events;
```

```
IF OBJECT_ID('Users', 'U') IS NOT NULL  
    DROP TABLE Users;
```

```
IF OBJECT_ID('Roles', 'U') IS NOT NULL  
    DROP TABLE Roles;
```

-- 1. ROLES TABLE

```
CREATE TABLE Roles( RoleId INT IDENTITY(1,1) PRIMARY KEY,  
                    RoleName VARCHAR(20) NOT NULL UNIQUE);
```



	TABLE_NAME
1	Roles
2	Users
3	Events
4	Categories
5	Enrolments
6	Results

-- 2. USERS TABLE

```
CREATE TABLE Users(UserId INT IDENTITY(1,1) PRIMARY KEY,
```

```
RoleId INT NOT NULL,Email VARCHAR(150) NOT NULL UNIQUE,  
PasswordHash VARCHAR(255) NOT NULL,  
FirstName VARCHAR(100) NOT NULL,  
LastName VARCHAR(100) NOT NULL,  
PhoneNumber VARCHAR(20) NULL,  
CreatedAt DATETIME2 NOT NULL DEFAULT GETDATE(),  
CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleId) REFERENCES Roles(RoleId));
```

-- 3. EVENTS TABLE

```
CREATE TABLE Events(EventId INT IDENTITY(1,1) PRIMARY KEY,OrganiserId INT NOT NULL,  
    EventName VARCHAR(200) NOT NULL,  
    Description VARCHAR(500) NOT NULL,EventDate DATETIME2 NOT NULL,  
    Location VARCHAR(200) NOT NULL,  
    Distance DECIMAL(6,2) NOT NULL,  
    EventType VARCHAR(20) NOT NULL,  
    CreatedAt DATETIME2 NOT NULL DEFAULT GETDATE(), CONSTRAINT FK_Events_Users  
    FOREIGN KEY (OrganiserId) REFERENCES Users(UserId),CONSTRAINT CK_Events_Distance  
    CHECK (Distance > 0),CONSTRAINT CK_Events_EventType
```

```
CHECK (EventType IN ('Run', 'Walk', 'Cycle')));
```

-- 4. CATEGORIES TABLE

```
CREATE TABLE Categories(CategoryId INT IDENTITY(1,1) PRIMARY KEY,  
    EventId INT NOT NULL,  
    CategoryName VARCHAR(100) NOT NULL,  
    AgeMin INT NULL,  
    AgeMax INT NULL,CONSTRAINT FK_Categories_Events FOREIGN KEY (EventId)  
    REFERENCES Events (EventId),  
    CONSTRAINT CK_Categories_AgeMin  
    CHECK (AgeMin IS NULL OR AgeMin >= 0),  
    CONSTRAINT CK_Categories_AgeMax  
    CHECK (AgeMax IS NULL OR AgeMax >= 0),  
    CONSTRAINT CK_Categories_AgeRange
```

```
CHECK(AgeMin IS NULL  
OR AgeMax IS NULL  
OR AgeMax >= AgeMin));
```

-- 5. ENROLMENTS TABLE

```
CREATE TABLE Enrolments (EnrolmentId INT IDENTITY(1,1) PRIMARY KEY,  
    ParticipantId INT NOT NULL,  
    CategoryId INT NOT NULL,  
    EnrolmentDate DATETIME2 NOT NULL DEFAULT GETDATE(),  
    CONSTRAINT FK_Enrolments_Users FOREIGN KEY (ParticipantId)  
    REFERENCES Users(UserId),CONSTRAINT FK_Enrolments_Categories  
    FOREIGN KEY (CategoryId) REFERENCES Categories(CategoryId),  
    CONSTRAINT UQ_Enrolments UNIQUE (ParticipantId, CategoryId));
```

-- 6. RESULTS TABLE

```
CREATE TABLE Results(ResultId INT IDENTITY(1,1) PRIMARY KEY, EnrolmentId INT NOT NULL,  
    FinishTime TIME NULL,Position INT NULL,  
    ResultStatus VARCHAR(20) NOT NULL DEFAULT 'Finished',
```

```
CONSTRAINT FK_Results_Enrolments FOREIGN KEY (EnrolmentId)
REFERENCES Enrolments(EnrolmentId),
CONSTRAINT UQ_Results_Enrolment UNIQUE (EnrolmentId),
CONSTRAINT CK_Results_Position CHECK (Position IS NULL OR Position > 0),
```

```
CONSTRAINT CK_Results_Status
CHECK (ResultStatus IN ('Finished', 'DNF', 'DNS', 'DQ')));
```

-- INSERT ROLES

```
INSERT INTO Roles (RoleName) VALUES
('Organiser'),
('Participant');
```

100 % 1 0 Ln: 30, Ch: 14 (20 chars, 1 lines) SPC CRLF UTF-8

	RoleId	RoleName
1	1	Organiser
2	2	Participant

Query executed successfully. localhost\SQLEXPRESS (16.0 ... CA14D03\linda (53) RaceDayDb 00:00:00 Row: 1, Col: 1 2 rows

-- INSERT USERS

```
INSERT INTO Users(RoleId,Email>PasswordHash,FirstName,LastName,PhoneNumber) VALUES
(1,'sipho.dlamini@raceday.co.za','PasswordHash1','Sipho','Dlamini','0821234567'),
(1,'anika.vandermerwe@raceday.co.za','PasswordHash2','Anika','Van Der Merwe','0839876543'),
(2,'thabo.mokoena@gmail.com','PasswordHash3','Thabo','Mokoena','0711112222'),
(2,'sarah.jenkins@gmail.com','PasswordHash4','Sarah','Jenkins','0723334444');
```

Results

Messages

	UserId	RoleId	Email	PasswordHash	FirstName	LastName	PhoneNumber	CreatedAt
1	1	1	sipho.dlamini@raceday.co.za	PasswordHash1	Sipho	Dlamini	0821234567	2026-09-04 14:20:31.1333333
2	2	1	anika.vandemerwe@raceday.co.za	PasswordHash2	Anika	Van Der Merwe	0839876543	2026-09-04 14:20:31.1333333
3	3	2	thabo.mokoena@gmail.com	PasswordHash3	Thabo	Mokoena	0711112222	2026-09-04 14:20:31.1333333
4	4	2	sarah.jenkins@gmail.com	PasswordHash4	Sarah	Jenkins	0723334444	2026-09-04 14:20:31.1333333

Query executed successfully.

localhost\SQLEXPRESS (16.0 ... | CA14D03\linda (53) | RaceDayDb | 00:00:00 | Row: 1, Col: 1 | 4 rows

Select Repository ^

Search

ENG US 14:25 2026/09/04

-- INSERT EVENTS

INSERT INTO Events(OrganiserId,EventName,Description,EventDate,Location,Distance,EventType)VALUES

(1,'Comrades Marathon 2027','A long-distance road running event between Pietermaritzburg and Durban.','2027-06-13 05:30:00',
'Durban, KwaZulu-Natal',89.00,'Run'),
(2,'Cape Town Cycle Tour 2027','A cycling event around the Cape Peninsula.','2027-03-14 06:00:00',
'Cape Town, Western Cape',109.00,'Cycle'),
(1,'Soweto Marathon 2027','A road running event through Soweto.','2027-11-07 05:30:00',
'Soweto, Gauteng',42.20,'Run');

Results		Messages							
	EventId	OrganiserId	EventName	Description	EventDate	Location	Distance	Event Type	
1	1	1	Comrades Marathon 2027	A long-distance road running event between Pieter...	2027-06-13 05:30:00.0000000	Durban, KwaZulu-Natal	89.00	Run	
2	2	2	Cape Town Cycle Tour 2027	A cycling event around the Cape Peninsula.	2027-03-14 06:00:00.0000000	Cape Town, Western Cape	109.00	Cycle	
3	3	1	Soweto Marathon 2027	A road running event through Soweto.	2027-11-07 05:30:00.0000000	Soweto, Gauteng	42.20	Run	

Query executed successfully. | localhost\SQLEXPRESS (16.0 ... | CA14D03\Linda (53) | RaceDayDb | 00:00:00 | Row: 1, Col: 1 | 3 rows

-- INSERT CATEGORIES

INSERT INTO Categories(EventId,CategoryName, AgeMin,AgeMax) VALUES

(1,'Open',20,39),

(1,'Veterans',40,49),

(2,'Main Race',18,75),

(2,'Short Route',12,80),

(3,'Full Marathon',20,70),

(3,'Half Marathon',16,75),
(3,'10km Community Run',10,85);

ResultsMessages

	CategoryId	EventId	CategoryName	AgeMin	AgeMax
1	1	1	Open	20	39
2	2	1	Veterans	40	49
3	3	2	Main Race	18	75
4	4	2	Short Route	12	80
5	5	3	Full Marathon	20	70
6	6	3	Half Marathon	16	75
7	7	3	10km Community Run	10	85

Query executed successfully.localhost\SQLEXPRESS (16.0 ...CA14D03\linda (53)RaceDayDb00:00:00Row: 1, Col: 17 rows

Select Repository ^

-- INSERT ENROLMENTS

INSERT INTO Enrolments(ParticipantId,CategoryId)VALUES

(3,1),

(3,5),

(4,3),

(4,6);

Results		Messages		
	EnrolmentId	ParticipantId	CategoryId	EnrolmentDate
1	1	3	1	2026-09-04 14:20:44.2166667
2	2	3	5	2026-09-04 14:20:44.2166667
3	3	4	3	2026-09-04 14:20:44.2166667
4	4	4	6	2026-09-04 14:20:44.2166667

Query executed successfully. | localhost\SQLEXPRESS (16.0 ...

-- INSERT RESULTS

```
INSERT INTO Results(EnrolmentId,FinishTime,Position,ResultStatus)VALUES
```

```
    (1,'06:45:12',412,'Finished'),
```

```
    (3,'03:15:40',105,'Finished');
```

Results		Messages			
	ResultId	EnrolmentId	FinishTime	Position	ResultStatus
1	1	1	06:45:12.0000000	412	Finished
2	2	3	03:15:40.0000000	105	Finished